

Data Governance Plan

RetailFlow Platform

Official Deliverable — Data Governance

Document purpose: present the data governance framework I designed for RetailFlow.

Document language: English.

Document scope: governance strategy, operating model, policies, standards, GDPR alignment, consent management, quality controls, auditability, AI governance, KPIs, change management and roadmap.

Updated version: this version integrates the latest RetailFlow implementation updates: final Streamlit consistency fixes, consent-aware AI display enforcement, AI Monitoring metrics based on [analytics_consent_count](#), Project Evidence skills matrix, Prometheus alert rules, Grafana dashboard evidence, CI/CD security reports, healthchecks, backup/restore scripts and read-only database role evidence.

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1. Executive Summary

RetailFlow is a recently established company that develops the **RetailFlow Platform**, a Retail Intelligence platform designed for e-commerce organizations.

The purpose of RetailFlow Platform is to transform customer events into trusted data, trusted data into customer intelligence, and customer intelligence into operational and strategic decision support.

Because RetailFlow is a new organization, I had the opportunity to design its data governance framework from the ground up.

Instead of adding governance after the platform had already grown, I designed RetailFlow according to a **Data Governance by Design** approach.

This means that data ownership, consent management, retention policies, quality controls, auditability, privacy principles and AI monitoring were integrated directly into the platform architecture.

The governance framework I defined covers:

- governance vision and scope;
- operating model and decision rights;
- roles and personas;
- consent management;
- data retention and anonymization;
- data quality controls;
- data classification;
- data security and access principles;
- auditability and evidence;
- AI governance;
- governance KPIs;
- risk management;
- change management;
- inclusion and accessibility;
- continuous improvement roadmap.

The governance framework is not only theoretical.

I implemented governance mechanisms directly in the platform through PostgreSQL schemas, FastAPI endpoints, Airflow DAGs, Streamlit governance dashboards and operational logs.

The most important governance implementation areas are:

Area	Implementation in RetailFlow
Consent management	Customer consent indicators, consent-aware analytics filtering and AI display blocking when analytics consent is absent
Data retention	Retention policy table and automated retention cleanup workflow
Anonymization	Customer anonymization logic and audit trail
Data quality	Validation rules, dead-letter events and quality logs
Auditability	Retention action logs, quality logs, dead-letter tables and operational evidence
AI governance	Consent-aware customer intelligence, model monitoring and drift reporting
Monitoring	Streamlit governance dashboard, FastAPI endpoints, Airflow workflows, Prometheus alert rules and Grafana dashboards

The result is a governance framework that is aligned with business value, technical implementation and regulatory expectations.

2. Governance Context

RetailFlow Platform is designed for an e-commerce context.

The platform collects and processes multiple categories of data:

- customer profiles;
- customer consent information;
- product catalog data;
- orders;
- payments;
- returns;
- shipments;
- sessions;
- product views;
- cart events;
- checkout events;
- support tickets;
- reviews;
- customer behavioral features;
- machine learning predictions;
- customer segments;
- data quality logs;
- retention and anonymization logs.

These data assets support different business and technical use cases:

- customer understanding;
- churn prevention;
- customer lifetime value estimation;

- segmentation;
- campaign prioritization;
- real-time event monitoring;
- ML performance monitoring;
- data quality management;
- compliance evidence;
- operational observability.

Because RetailFlow uses customer-level data and derived AI outputs, governance is a core platform requirement.

The main governance challenge is to ensure that data is:

- trusted;
- compliant;
- traceable;
- secure;
- usable;
- auditable;
- understandable;
- aligned with business purpose.

Without governance, the platform could expose the company to several risks:

- customer data misuse;
- analytics performed without consent;
- uncontrolled data retention;
- poor data quality affecting AI outputs;
- lack of accountability;
- unclear data ownership;
- inability to explain or audit decisions;
- model monitoring gaps;
- operational blind spots.

I therefore designed the governance layer as an integrated part of the platform rather than as a separate administrative document.

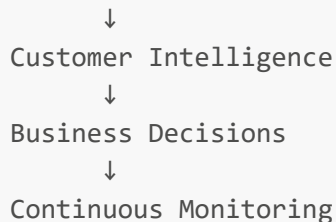
3. Governance Vision

The governance vision for RetailFlow is:

I designed RetailFlow governance to ensure that customer data can be used as a trusted, compliant and valuable asset while preserving privacy, accountability, quality and auditability across the platform.

This vision supports the broader product vision:

Customer Events
↓
Trusted Data



The governance layer ensures that each step of this value chain remains controlled.

Value chain step	Governance contribution
Customer Events	Validation rules and dead-letter handling
Trusted Data	Data quality controls and schema organization
Customer Intelligence	Consent-aware analytics and AI governance
Business Decisions	Clear definitions, roles and quality KPIs
Continuous Monitoring	Audit logs, dashboards and governance KPIs

The key principle is simple:

Data should only become intelligence if it is governed, reliable, traceable and used for a legitimate purpose.

4. Governance Objectives

I defined the following governance objectives for RetailFlow.

4.1 Ensure clear data ownership

Each critical data domain must have an accountable owner.

This avoids ambiguity around:

- who validates definitions;
- who approves business usage;
- who owns data quality targets;
- who resolves business conflicts;
- who decides whether a data product is fit for use.

4.2 Protect customer data

RetailFlow processes customer-related data.

I therefore integrated privacy principles such as:

- purpose limitation;
- consent management;
- minimization;
- storage limitation;

- anonymization;
- accountability;
- auditability.

4.3 Make analytics consent-aware

Customer intelligence can influence marketing, retention and personalization decisions.

For this reason, I implemented a consent-aware analytics principle:

```
Customer-level AI predictions must only be displayed when analytics consent is granted.
```

This is visible in the Customer Intelligence interface.

The user can filter customers by analytics consent.

If the filter is disabled for governance demonstration purposes and a customer without analytics consent is selected, the platform does not display churn, CLV, segmentation or AI recommendations.

Instead, Streamlit displays a governance message explaining that AI predictions are not available because analytics consent was not granted.

This makes the consent rule observable and testable inside the application rather than only documented as a policy.

4.4 Control data quality

Invalid events must not silently enter trusted analytical tables.

I implemented a pipeline quality strategy based on:

- validation rules;
- rejection logic;
- dead-letter events;
- quality logs;
- monitoring dashboards.

4.5 Ensure auditability

Governance must produce evidence.

I therefore designed and implemented audit traces for:

- retention actions;
- anonymization actions;
- dead-letter events;
- data quality checks;
- ML reports;
- Airflow workflows;

- platform monitoring.

4.6 Govern the AI lifecycle

RetailFlow includes churn, CLV and segmentation models.

I included AI governance principles covering:

- model purpose;
- data eligibility;
- explainability;
- monitoring;
- drift detection;
- retraining;
- human oversight;
- responsible use of predictions.

4.7 Support business adoption

Governance should enable decision-making rather than slow down the business.

I therefore designed governance as:

- practical;
- role-based;
- measurable;
- integrated into tools;
- visible through dashboards;
- connected to business value.

5. Governance Scope

The governance scope is focused on the data domains that are most critical for RetailFlow Platform.

5.1 In scope

The following areas are in scope.

Domain	Included assets
Customer Data	Customer profile, consent, account status, anonymization status
Behavioral Events	Product views, cart events, checkout events, purchase events
Transaction Data	Orders, order items, payments, returns, shipments
Product Data	Product catalog, categories, supplier references
Customer Features	Behavioral and transactional aggregates
AI Outputs	Churn predictions, CLV predictions, customer segments

Domain	Included assets
Governance Data	Consent records, retention policies, retention action logs
Quality Data	Dead-letter events and data quality logs
Monitoring Data	Platform health, API metrics, PostgreSQL metrics, Airflow health

5.2 Out of scope

The following areas are not covered by the current governance scope:

- Enterprise Identity and Access Management;
- Single Sign-On;
- multi-region deployment;
- 24/7 production support and on-call operations.

These areas are identified as future enterprise-level capabilities.

5.3 Scope rationale

I deliberately focused the governance scope on the most valuable and risky domains first.

Customer data, behavioral events and AI outputs are the most important areas because they directly affect:

- customer privacy;
- business decision-making;
- ML reliability;
- retention actions;
- customer segmentation;
- marketing activation;
- compliance obligations.

This phased governance scope avoids trying to govern everything at once while still covering the most critical platform risks.

6. Governance by Design

RetailFlow was designed as a new platform.

This created an opportunity to implement governance from the beginning rather than adding it later.

I used a **Governance by Design** approach.

This means that governance principles are integrated into the platform architecture, data model, pipelines, dashboards and operational workflows.

6.1 Governance by Design principles

Principle	Implementation in RetailFlow
Privacy by Design	Consent fields, anonymization logic and retention policies are part of the data model.
Quality by Design	Event validation happens before event persistence.
Auditability by Design	Retention actions and quality issues are logged.
AI Governance by Design	ML predictions are monitored and linked to consent-aware exploration.
Observability by Design	Platform health and metrics are exposed through Prometheus, Grafana and Streamlit.
Accountability by Design	Roles and personas are defined for ownership, stewardship and compliance.

6.2 Governance maturity

Although RetailFlow is a recently established organization, its governance maturity is relatively high because governance was integrated early.

I evaluate the current maturity as follows:

Dimension	Current maturity	Explanation
Data ownership	Advanced	Roles and personas are defined by domain.
Consent management	Advanced	Consent flags are integrated into customer data and analytics usage.
Retention and anonymization	Advanced	Retention policies and anonymization workflow are implemented.
Auditability	Advanced	Logs exist for retention actions and quality issues.
Data quality	Advanced	Validation, dead-letter events and quality dashboards exist.
AI governance	Intermediate to Advanced	ML metrics, drift and explainability are implemented.
Metadata management	Developing	Technical metadata exists, but cataloging can be improved.
Enterprise data catalog	Planned	A full data catalog is a future improvement.
Access management	Developing	Future role-based access control can strengthen governance.

This maturity profile is realistic.

It recognizes that the project already contains strong governance controls while also identifying areas that would need to be expanded in an enterprise environment.

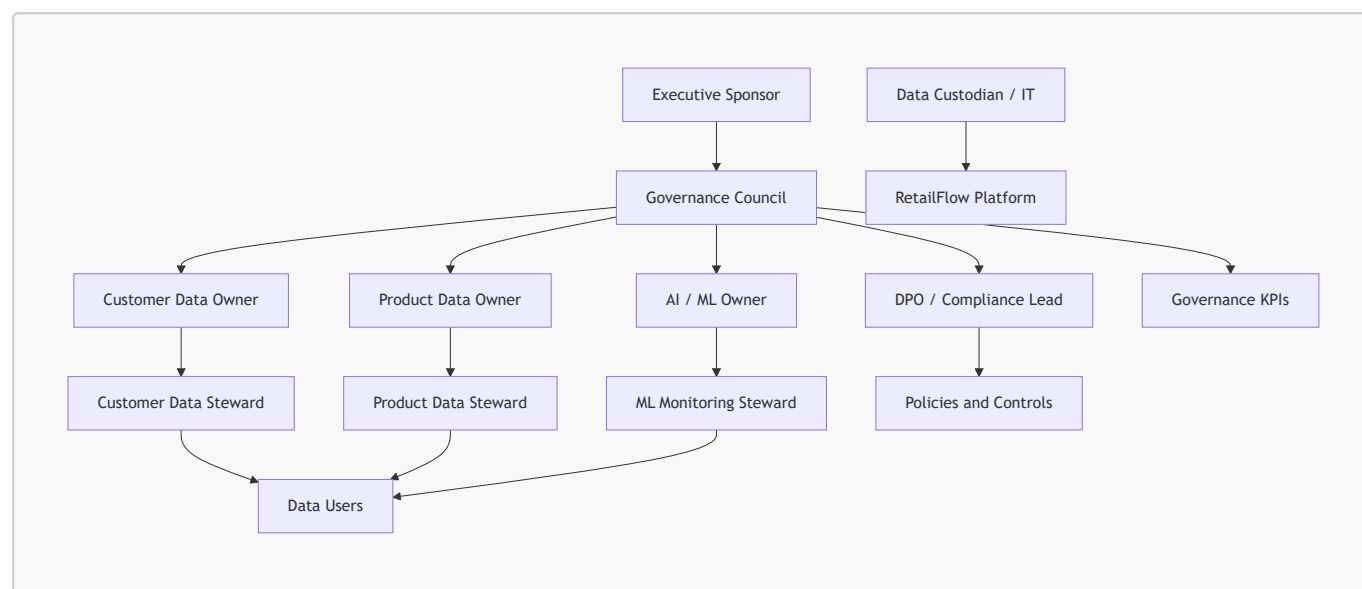
7. Operating Model

I designed a hybrid governance operating model.

The central governance layer defines common policies, standards and controls.

Domain-specific actors apply these rules in their business or technical areas.

7.1 Hybrid governance model



7.2 Why hybrid governance

A fully centralized model would be too rigid for a platform that includes business, engineering and AI topics.

A fully decentralized model would create inconsistent definitions and weak accountability.

The hybrid model provides:

- central consistency;
- domain accountability;
- faster operational execution;
- clear escalation paths;
- scalable governance practices.

7.3 Governance cadence

I defined the following governance cadence.

Activity	Frequency	Responsible role
Governance KPI review	Monthly	Governance Council
Data quality issue review	Weekly	Data Steward
Retention action review	Monthly	DPO / Compliance Lead

Activity	Frequency	Responsible role
ML monitoring review	Monthly	ML Owner
Risk register review	Quarterly	Governance Council
Policy review	Quarterly	Governance Council
Accessibility and training review	Quarterly	Executive Sponsor and Governance Council

8. Roles, Responsibilities and Personas

To make the governance model concrete, I associated each governance role with a realistic RetailFlow persona.

These personas are not necessarily separate full-time positions.

They can be responsibilities assigned to existing job roles.

8.1 Role and persona mapping

Governance role	RetailFlow persona	Main responsibility
Executive Sponsor	Chief Data & Analytics Officer	Sponsors the data governance program and validates strategic priorities.
Governance Council	Cross-functional governance committee	Approves policies, reviews risks and arbitrates decisions.
Data Owner	Head of Customer Intelligence	Owns customer analytics definitions, usage priorities and quality targets.
Data Steward	Senior CRM & Analytics Manager	Monitors quality, glossary definitions, consent usage and issue resolution.
Data Custodian	Lead Data Engineer	Operates data systems, pipelines, security controls and retention workflows.
DPO / Compliance Lead	Privacy & Compliance Manager	Oversees GDPR alignment, consent, retention and audit readiness.
ML Owner	Lead Machine Learning Engineer	Owns model monitoring, drift analysis, retraining and explainability.
Business Owner	Head of E-Commerce Performance	Uses customer intelligence for business decisions and campaign prioritization.
Data Users	Marketing analysts, CRM specialists, business analysts	Consume governed data and report quality issues.

8.2 Executive Sponsor

Persona: Chief Data & Analytics Officer.

The Executive Sponsor provides authority, funding and visibility.

Responsibilities:

- approve the governance strategy;
- validate priorities;
- remove cross-functional blockers;
- sponsor the governance roadmap;
- ensure alignment with business objectives;
- review governance maturity.

8.3 Governance Council

Persona: Monthly cross-functional committee.

Typical members:

- Chief Data & Analytics Officer;
- Head of Customer Intelligence;
- Privacy & Compliance Manager;
- Lead Data Engineer;
- Lead Machine Learning Engineer;
- Head of E-Commerce Performance;
- Senior CRM & Analytics Manager.

Responsibilities:

- approve governance policies;
- define common standards;
- review KPIs;
- review risks;
- resolve ownership conflicts;
- prioritize governance improvements;
- validate change management actions.

8.4 Data Owner

Persona: Head of Customer Intelligence.

The Data Owner is accountable for the business meaning and usage of customer intelligence data.

Responsibilities:

- define business definitions;
- approve customer analytics use cases;
- validate quality targets;
- prioritize customer data improvements;
- arbitrate business conflicts around metrics;
- ensure the domain produces value.

8.5 Data Steward

Persona: Senior CRM & Analytics Manager.

The Data Steward manages governance in daily operations.

Responsibilities:

- monitor quality indicators;
- maintain glossary definitions;
- review data quality issues;
- validate consent usage practices;
- coordinate issue resolution;
- communicate governance rules to users.

8.6 Data Custodian

Persona: Lead Data Engineer.

The Data Custodian implements governance controls technically.

Responsibilities:

- operate PostgreSQL schemas;
- maintain pipelines;
- implement validation rules;
- maintain Airflow workflows;
- implement retention cleanup;
- maintain quality and audit logs;
- support monitoring and observability.

8.7 DPO / Compliance Lead

Persona: Privacy & Compliance Manager.

The DPO / Compliance Lead oversees privacy and regulatory alignment.

Responsibilities:

- validate GDPR alignment;
- review consent management;
- review retention policies;
- oversee anonymization logic;
- ensure audit evidence is available;
- review privacy risk controls;
- validate training material for privacy awareness.

8.8 ML Owner

Persona: Lead Machine Learning Engineer.

The ML Owner governs the AI lifecycle.

Responsibilities:

- validate model purpose;

- monitor model metrics;
- review feature importance;
- monitor drift reports;
- validate retraining workflows;
- ensure model outputs are explainable;
- document responsible use of predictions.

8.9 Business Owner

Persona: Head of E-Commerce Performance.

The Business Owner ensures that governed data supports decision-making.

Responsibilities:

- use churn, CLV and segmentation outputs;
- prioritize business actions;
- validate business usefulness;
- provide feedback on dashboards;
- ensure insights are used responsibly.

8.10 Data Users

Personas: Marketing analysts, CRM specialists and business analysts.

Data Users consume governed data under approved rules.

Responsibilities:

- follow data usage policies;
- respect consent constraints;
- use approved definitions;
- report data quality issues;
- request clarification when definitions are unclear;
- participate in training.

9. Governance Decision Model

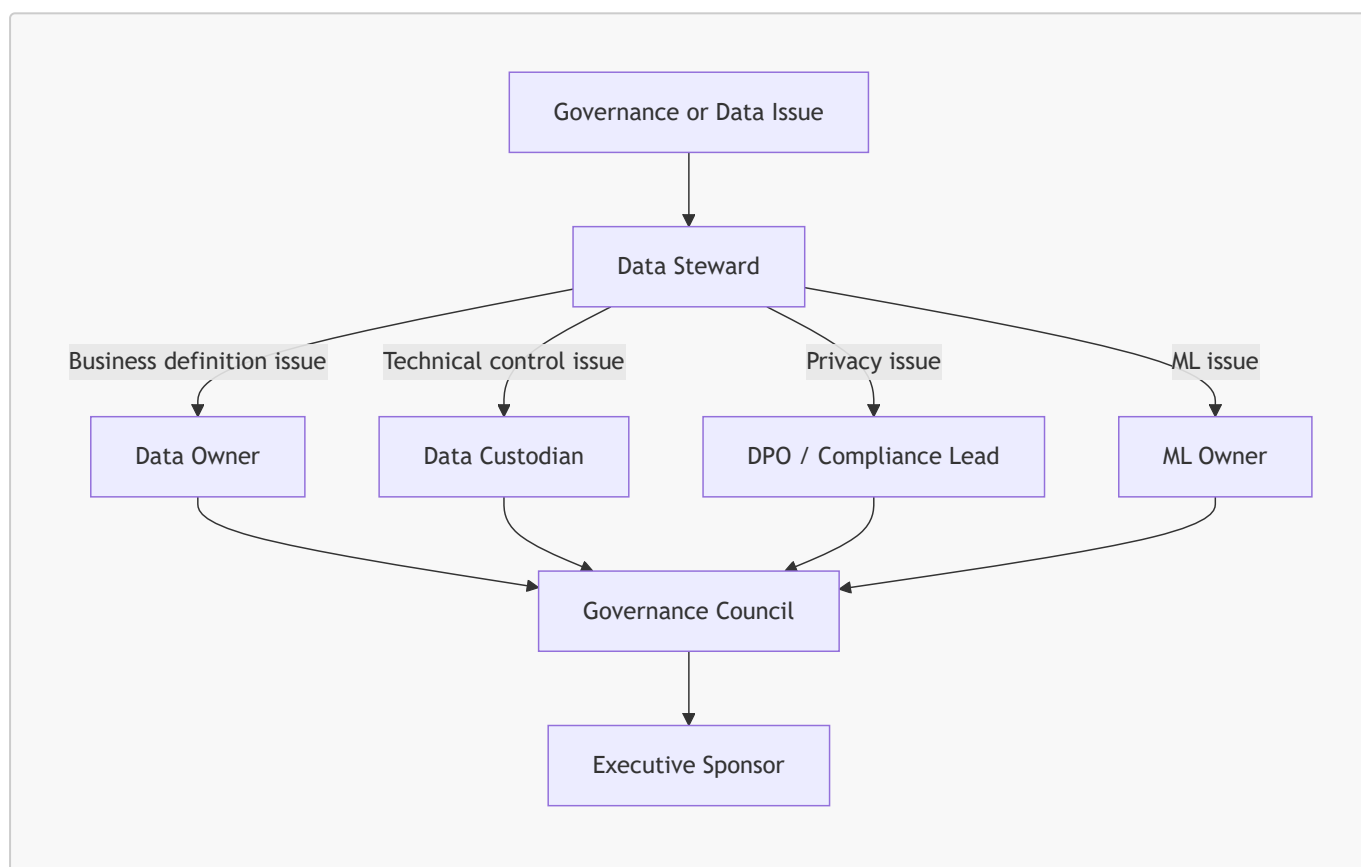
I defined decision rights to clarify who decides what.

9.1 Decision rights matrix

Decision area	Decision owner	Consulted roles	Evidence required
New customer analytics use case	Data Owner	DPO, ML Owner, Business Owner	Purpose, data fields, consent requirement
New data quality rule	Data Steward	Data Custodian, Data Owner	Rule definition, severity, remediation path

Decision area	Decision owner	Consulted roles	Evidence required
Retention policy update	DPO / Compliance Lead	Data Owner, Data Custodian	Legal basis, target table, action
Model retraining approval	ML Owner	Data Owner, Data Custodian	Metrics, drift report, validation output
Glossary definition update	Data Steward	Data Owner, Data Users	Definition, examples, owner approval
Governance KPI target update	Governance Council	Executive Sponsor	KPI history, business impact
Access policy update	Governance Council	DPO, Data Custodian	Role mapping and security impact

9.2 Escalation path



10. Data Domains Under Governance

I structured governance around core domains.

10.1 Customer domain

The customer domain includes:

- customer identifier;
- demographic attributes;

- account status;
- loyalty status;
- consent indicators;
- anonymization status;
- last interaction timestamp.

Governance focus:

- privacy;
- consent;
- retention;
- segmentation eligibility;
- analytics usage;
- anonymization.

10.2 Event domain

The event domain includes:

- product views;
- add-to-cart events;
- checkout events;
- purchase events;
- session events;
- raw payloads.

Governance focus:

- validation;
- traceability;
- timeliness;
- dead-letter handling;
- quality monitoring.

10.3 Transaction domain

The transaction domain includes:

- orders;
- order items;
- payments;
- shipments;
- returns;
- refunds.

Governance focus:

- accuracy;
- financial consistency;
- reporting reliability;

- retention requirements.

10.4 Product domain

The product domain includes:

- product catalog;
- category;
- supplier;
- product availability;
- price;
- product metadata.

Governance focus:

- completeness;
- consistency;
- reference values;
- product recommendations;
- catalog quality.

10.5 AI output domain

The AI output domain includes:

- churn scores;
- churn risk labels;
- CLV predictions;
- CLV value bands;
- customer segments;
- model versions;
- prediction timestamps;
- drift metrics.

Governance focus:

- explainability;
- monitoring;
- model versioning;
- responsible use;
- consent-aware exploration;
- retraining.

11. Data Classification

I defined a simple classification model adapted to RetailFlow.

The goal is to classify data by sensitivity and apply appropriate controls.

11.1 Classification levels

Classification	Description	Examples	Governance controls
Public	Information that can be shared externally without customer risk.	Product catalog, product category labels, generic platform description	Basic integrity controls
Internal	Operational or analytical information intended for internal use.	Aggregated sales KPIs, model monitoring summaries, operational metrics	Internal access rules and documentation
Confidential	Customer-related or business-sensitive data requiring stronger controls.	Customer profiles, consent flags, transaction history, behavioral features, ML predictions	Consent rules, retention, audit logs, limited access

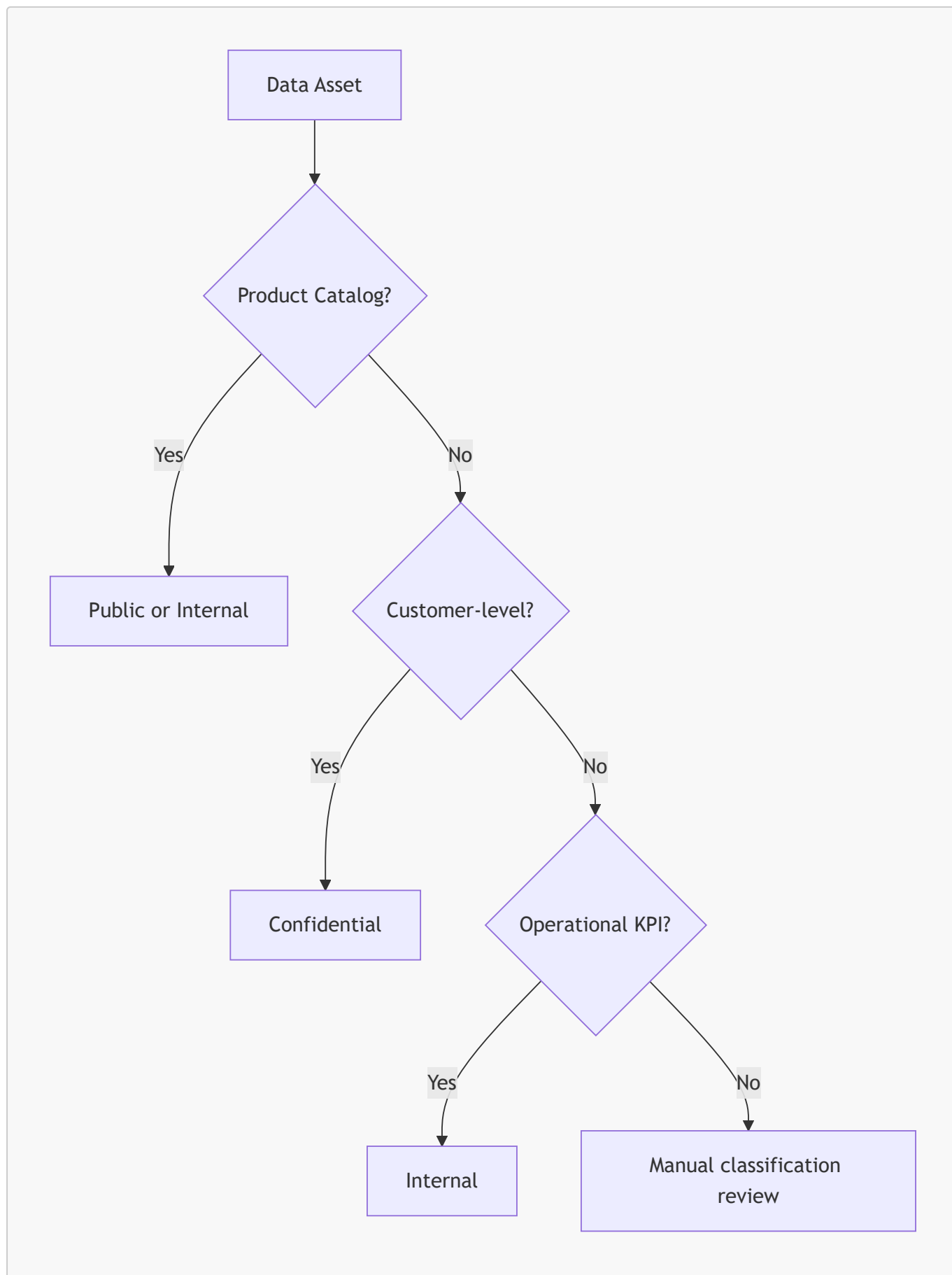
11.2 Classification by data domain

Data domain	Classification	Reason
Product catalog	Public / Internal	Product information can be public, but internal pricing or margin can be sensitive.
Customer profiles	Confidential	Contains customer-level information.
Consent data	Confidential	Directly related to privacy rights and permitted usage.
Orders and payments	Confidential	Transactional and potentially sensitive business data.
Behavioral events	Confidential	Can describe individual customer behavior.
Customer features	Confidential	Derived customer-level analytical data.
ML predictions	Confidential	Can influence customer treatment and marketing actions.
Aggregated KPIs	Internal	Business performance metrics should remain internal.
Monitoring metrics	Internal	Operational information for platform teams.
Public documentation	Public	Non-sensitive product and architecture descriptions.

11.3 Handling rules

Classification	Handling rules
Public	Can be documented and shared externally if validated.
Internal	Accessible to approved internal users only.
Confidential	Requires purpose, role, consent consideration, retention rule and auditability.

11.4 Classification diagram



12. Data Policies

I defined governance policies that translate the governance principles into operational rules.

12.1 Policy overview

Policy	Purpose
Customer data policy	Define how customer data may be used.
Consent management policy	Control marketing, analytics and personalization usage.
Retention policy	Define how long data is retained and what action is applied.
Anonymization policy	Define how customer identity data is removed or neutralized.
Data quality policy	Define validation rules and issue management.
AI governance policy	Define responsible use of ML predictions.
Access and security policy	Define access principles and technical controls.
Audit policy	Define evidence, logging and review expectations.

12.2 Customer data policy

Customer data may only be used for approved purposes.

Approved purposes include:

- customer intelligence;
- service improvement;
- churn prevention;
- segmentation;
- CLV analysis;
- operational monitoring;
- data quality analysis;
- compliance and retention processes.

Customer data must not be used for undefined purposes without review.

The Data Owner and DPO / Compliance Lead must be consulted for any new customer-level analytical use case.

12.3 Acceptable use policy

Users consuming RetailFlow data must:

- use approved dashboards and endpoints;
- respect consent indicators;
- avoid exporting unnecessary customer-level data;
- use aggregated data where possible;
- report quality issues;
- follow glossary definitions;
- avoid unsupported interpretations of AI outputs.

12.4 Policy review

Policies must be reviewed quarterly by the Governance Council.

Policy updates should be triggered when:

- new data domains are added;
- new ML use cases are introduced;
- a privacy risk is identified;
- recurring quality issues appear;
- new compliance requirements emerge;
- business usage changes.

13. Consent Management Policy

Consent management is central to RetailFlow governance.

The platform tracks consent at customer level.

13.1 Consent dimensions

Consent field	Purpose
marketing_consent	Indicates whether the customer can be targeted for marketing activation.
analytics_consent	Indicates whether the customer can be used in analytics and customer intelligence exploration.
personalization_consent	Indicates whether the customer can be used for personalization-related use cases.

13.2 Consent usage principles

I defined the following principles:

1. Consent must be explicit enough to support the intended purpose.
2. Customer-level AI predictions should be displayed only when analytics consent is granted.
3. Marketing activation should require marketing consent.
4. Personalization use cases should require personalization consent.
5. Consent values must be visible to data users where they affect usage.
6. Consent must be checked before using or displaying AI outputs for customer-level actions.

13.3 Consent-aware analytics

RetailFlow connects governance to analytics through the Customer Intelligence page.

The customer explorer contains a default filter:

Show only customers with analytics consent

This filter is enabled by default.

The page also implements a stronger governance rule.

When a customer does not have analytics consent, RetailFlow hides customer-level AI outputs from the interface:

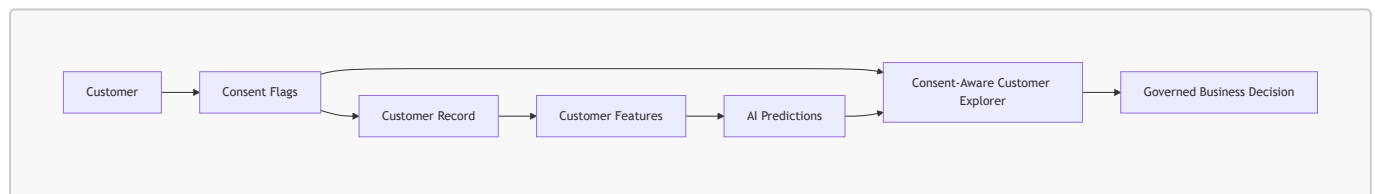
- churn prediction;
- CLV prediction;
- segmentation result;
- AI-driven recommended actions;
- raw AI profile.

The page displays the following business-readable governance message:

Ce client n'a pas donné son consentement analytics. Les prédictions IA ne sont donc pas disponibles.

This demonstrates that analytics consent is enforced at the application display layer and not only stored in the database.

13.4 Consent flow



13.5 Consent monitoring

Consent rates are visible in the Data Governance dashboard.

The dashboard tracks:

- marketing consent rate;
- analytics consent rate;
- personalization consent rate;
- anonymized customers;
- customer count.

The AI Monitoring page also uses `analytics_consent_count` as the visible count for AI-authorized customer outputs.

This means the AI monitoring view is aligned with the same governance rule as the Customer Intelligence dashboard.

This makes consent not only stored, but also monitored and connected to AI visibility.

13.6 Current consent implementation evidence

Evidence	Implementation	Where to verify
Customer consent storage	<code>marketing_consent</code> , <code>analytics_consent</code> , <code>personalization_consent</code>	PostgreSQL <code>core.customers</code> and governance dashboards
Governance summary	Consent counts exposed by API	FastAPI <code>GET /governance/summary</code>
Consent-aware customer list	Analytics consent filter in customer API	FastAPI <code>GET /ai/customers?analytics_consent_only=true</code>
AI display control	Churn, CLV, segmentation and recommendations hidden when analytics consent is absent	Streamlit <code>Customer Intelligence</code>
AI monitoring count	<code>analytics_consent_count</code> used for visible AI-authorized customer outputs	Streamlit <code>AI Monitoring</code>

14. Retention and Anonymization Policy

RetailFlow includes a retention policy framework and automated anonymization workflow.

14.1 Retention policy table

Retention policies are stored in:

```
governance.data_retention_policies
```

This table defines:

- policy identifier;
- target domain;
- target table;
- retention duration;
- retention action;
- owner role;
- legal basis or governance basis.

14.2 Retention principles

I defined the following retention principles:

1. Data should not be retained indefinitely without purpose.
2. Customer personal data should be anonymized when the retention condition is met.
3. Retention actions must be logged.
4. The retention workflow must be auditable.
5. Retention policies must be reviewed periodically.

14.3 Anonymization workflow

The Airflow DAG `retention_cleanup` supports the retention process.

The workflow:

1. identifies customers affected by the retention policy;
2. anonymizes personal fields;
3. disables consent flags;
4. changes account status to anonymized;
5. records the action in the retention audit log.

14.4 Anonymized fields

The anonymization process affects fields such as:

- first name;
- last name;
- email;
- phone number;
- birth date;
- gender;
- city;
- postal code;
- consent flags;
- account status.

14.5 Retention action log

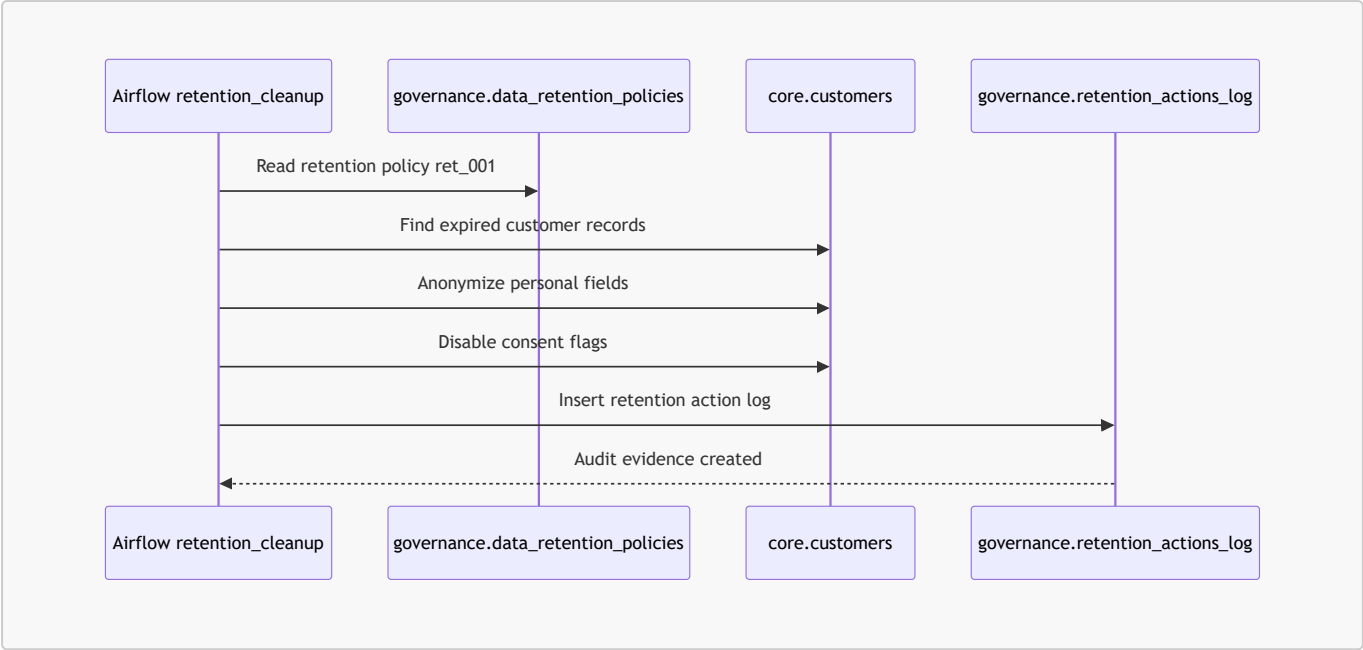
Retention actions are logged in:

```
governance.retention_actions_log
```

The log stores:

- action identifier;
- policy identifier;
- table name;
- record identifier;
- action type;
- action status;
- execution timestamp;
- executing component;
- action details.

14.6 Retention workflow diagram



15. Data Quality Policy

Data quality is a governance requirement in RetailFlow.

The objective is to prevent incorrect data from contaminating downstream analytics and ML outputs.

15.1 Quality dimensions

I defined quality controls around the following dimensions:

Dimension	Meaning in RetailFlow
Completeness	Required fields must be present.
Validity	Event types and values must be allowed.
Consistency	Events must reference existing customers and products.
Timeliness	Timestamps must be valid and usable.
Traceability	Errors must be logged and explainable.

15.2 Quality validation rules

The real-time pipeline includes validation rules such as:

Rule	Purpose	Action
Event identifier required	Ensure event traceability	Reject invalid event
Event type allowed	Prevent unsupported event categories	Reject invalid event
Customer exists	Ensure customer referential integrity	Reject invalid event
Product exists	Ensure product referential integrity	Reject invalid event

Rule	Purpose	Action
Timestamp valid	Ensure event chronology is usable	Reject invalid event

15.3 Dead-letter handling

Invalid events are stored in:

```
governance.dead_letter_events
```

This prevents invalid data from entering trusted analytical tables.

15.4 Quality logs

Failed rules are logged in:

```
governance.data_quality_logs
```

This allows quality issues to be:

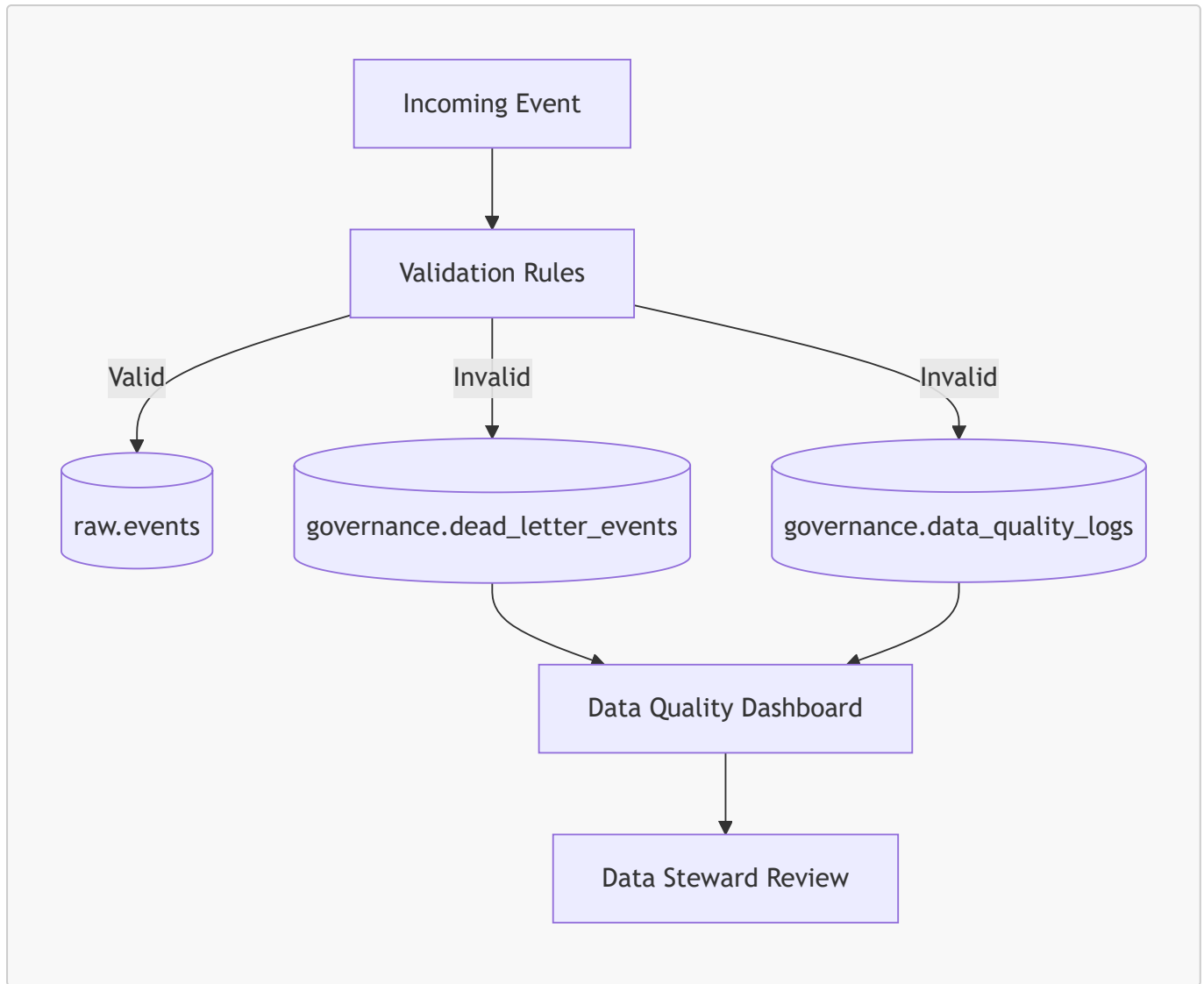
- counted;
- reviewed;
- categorized;
- assigned;
- monitored;
- audited.

15.5 Quality monitoring

The Data Quality dashboard displays:

- dead-letter events;
- failed rules;
- severity distribution;
- impacted event types;
- quality rule summaries;
- technical evidence.

15.6 Quality workflow diagram



16. Data Security and Access Policy

RetailFlow is not currently positioned as a complete enterprise IAM platform.

However, I defined security principles that guide data access and future implementation.

16.1 Security principles

Principle	Description
Least privilege	Users should access only what they need.
Purpose-based access	Access should be linked to a business purpose.
Confidentiality	Customer-level data should be protected.
Auditability	Sensitive actions should be traceable.
Segregation of duties	Governance decisions and technical implementation should not be controlled by one person only.
Secure defaults	Sensitive data should not be exposed by default.

16.2 Access expectations by role

Role	Expected access
Data Owner	Domain-level KPIs, definitions and governance reports
Data Steward	Quality logs, glossary, consent indicators and issue tracking
Data Custodian	Technical schemas, pipelines, logs and platform operations
DPO / Compliance Lead	Consent, retention, anonymization and audit trails
ML Owner	Features, model reports, prediction summaries and drift outputs
Business User	Approved dashboards and aggregated insights

16.3 Future access improvements

Future access improvements should include:

- role-based access control;
- authentication;
- API authorization scopes;
- audit logging for user-level access;
- secrets management;
- masking for sensitive attributes;
- stronger separation between user personas.

17. AI Governance Policy

RetailFlow includes AI outputs that can influence business decisions.

I therefore designed a specific AI governance policy.

17.1 Governed AI use cases

Model	Use case	Governance concern
Churn model	Identify customers at risk	Avoid over-automated treatment and ensure explainability
CLV model	Estimate customer value	Avoid unfair resource allocation without business review
Segmentation model	Group customers	Ensure segments are understandable and actionable

17.2 AI governance principles

I defined the following principles:

1. AI predictions must support decisions, not replace human judgment.

2. Customer-level AI exploration must enforce analytics consent before displaying predictions.
3. Models must be monitored with metrics and drift signals.
4. Model outputs must be explainable to business users.
5. Retraining must be scheduled and traceable.
6. AI outputs must be versioned and timestamped.
7. Business users must understand the limits of model predictions.
8. AI dashboards must make the consent boundary visible and testable.

17.3 Model monitoring

RetailFlow monitors:

- churn ROC AUC;
- churn F1;
- churn precision;
- churn recall;
- Brier score;
- CLV MAE;
- CLV RMSE;
- CLV R²;
- segmentation quality;
- feature importance;
- prediction distribution;
- drift status.

17.4 Drift monitoring

Drift monitoring detects changes in customer behavior that may reduce model reliability.

The AI Monitoring page displays drift status and drifted feature count.

17.5 Retraining governance

The Airflow DAG `ml_retraining` orchestrates:

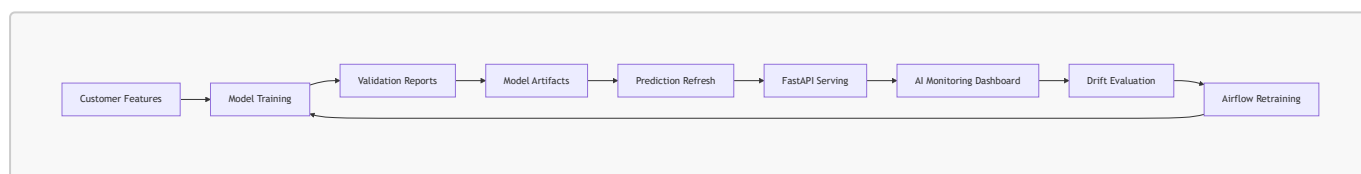
- churn model training;
- segmentation training;
- CLV model training;
- prediction refresh;
- drift evaluation.

17.6 Current AI governance implementation evidence

Control	Current RetailFlow implementation	Evidence location
Consent-aware AI display	Customer-level AI outputs are hidden when <code>analytics_consent = false</code> .	Streamlit <code>Customer Intelligence</code>

Control	Current RetailFlow implementation	Evidence location
AI-authorized customer count	AI Monitoring uses <code>analytics_consent_count</code> to represent visible AI-authorized customer outputs.	Streamlit <code>AI Monitoring</code>
Human-readable actioning	Decision framework translates churn, CLV and segment outputs into business recommendations.	Streamlit <code>Customer Intelligence</code>
Model registry	Model registry is generated and displayed.	<code>ml/model_registry.json</code> , Streamlit <code>AI Monitoring</code>
Model reports	Churn, CLV, segmentation, model summary and drift reports are displayed.	<code>ml/reports/</code> , Streamlit <code>AI Monitoring</code>
Retraining traceability	Retraining runs are logged and exposed in the monitoring page.	<code>ml/reports/retraining_runs.json</code> , Airflow <code>ml_retraining</code>
CI/CD validation	Tests and security checks run in GitHub Actions.	GitHub Actions, <code>.github/workflows/ci.yml</code>

17.6 AI governance lifecycle



18. Business Glossary

I defined a business glossary to clarify key terms used in RetailFlow.

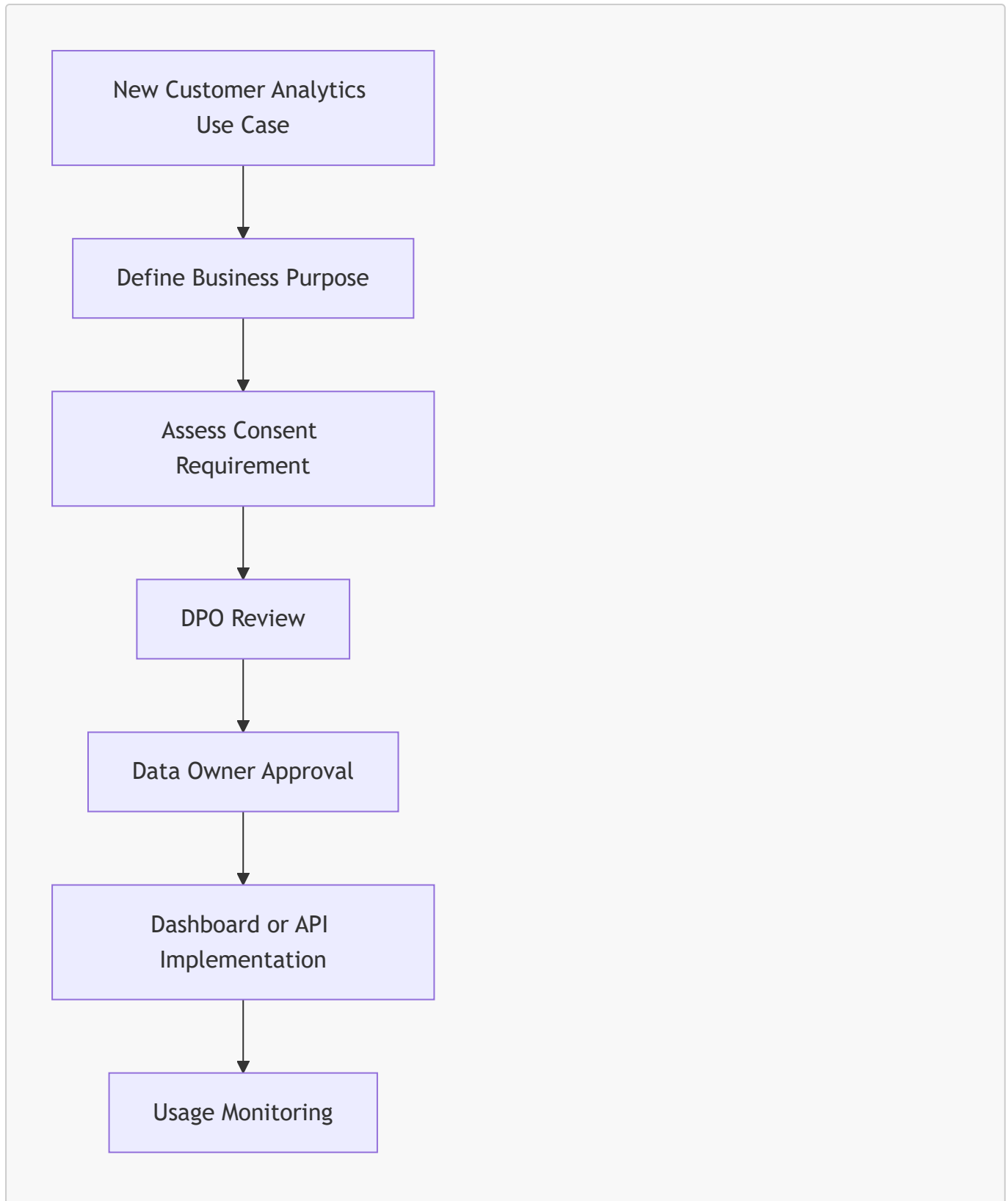
Term	Definition	Owner
Active customer	A customer with recent purchase or behavioral activity in the platform.	Head of Customer Intelligence
Analytics consent	Permission indicator allowing customer data to be used for analytics and customer intelligence exploration.	DPO / Compliance Lead
Marketing consent	Permission indicator allowing marketing activation.	DPO / Compliance Lead
Personalization consent	Permission indicator allowing personalized recommendations or experiences.	DPO / Compliance Lead
Customer event	A behavioral action generated by a customer, such as product view, cart or checkout activity.	Data Steward
Valid event	An event that passes required validation rules before persistence.	Data Custodian

Term	Definition	Owner
Dead-letter event	An event rejected by validation and isolated for review.	Data Steward
Churn risk	Probability or label indicating the likelihood of customer disengagement.	ML Owner
CLV	Customer Lifetime Value, the estimated future value of a customer.	Head of Customer Intelligence
Customer segment	Business-readable group of customers with similar behavior or value profile.	ML Owner
Retention policy	Rule defining how long a data asset is kept and what action is applied.	DPO / Compliance Lead
Anonymization	Process of removing or neutralizing identifying customer attributes.	DPO / Compliance Lead
Data quality rule	A rule used to validate data completeness, validity or consistency.	Data Steward
Drift	A change in data distribution that can affect model reliability.	ML Owner
Audit trail	Recorded evidence of governance-related actions.	Governance Council

19. Governance Processes

I defined governance processes to make policies operational.

19.1 Consent review process



19.2 Data quality issue process

1. Data quality issue is detected.
2. Issue is logged in the governance schema.
3. Data Steward reviews severity.
4. Data Custodian investigates technical cause.
5. Data Owner validates business impact.
6. Corrective action is implemented.
7. Issue is monitored for recurrence.

19.3 Retention process

1. Retention policy is defined.
2. Airflow retention workflow identifies eligible records.
3. Customer record is anonymized.
4. Consent flags are disabled.
5. Retention action is logged.
6. Governance dashboard exposes the action.
7. Compliance Lead reviews the audit trail.

19.4 AI monitoring process

1. Models are trained.
2. Predictions are refreshed.
3. Metrics are generated.
4. Drift is evaluated.
5. AI Monitoring dashboard displays outputs.
6. ML Owner reviews model status.
7. Retraining or investigation is triggered if needed.

20. Data Quality Controls

Data quality controls protect the platform from poor input data.

20.1 Preventive controls

Control	Description
API schemas	Incoming requests must follow expected structures.
Event validation	The consumer validates business and technical rules.
Referential checks	Customer and product identifiers are checked.
Allowed event types	Unsupported event types are rejected.
Timestamp validation	Invalid timestamps are rejected.

20.2 Detective controls

Control	Description
Dead-letter monitoring	Rejected events are visible in dashboards.
Quality summaries	Failed rules are counted and reviewed.
Airflow data quality DAG	Quality checks are scheduled.
Streamlit Data Quality page	Quality issues are exposed to users.

20.3 Corrective controls

Control	Description
Dead-letter review	Data Steward reviews rejected events.
Rule adjustment	Data quality rules can be adjusted if needed.
Pipeline correction	Data Custodian fixes technical causes.
Reprocessing path	Future improvement for corrected events.

21. Metadata, Lineage and Traceability

Metadata and lineage are important for trust.

RetailFlow currently implements practical traceability through schemas, logs and dashboards.

21.1 Technical metadata

Technical metadata exists through:

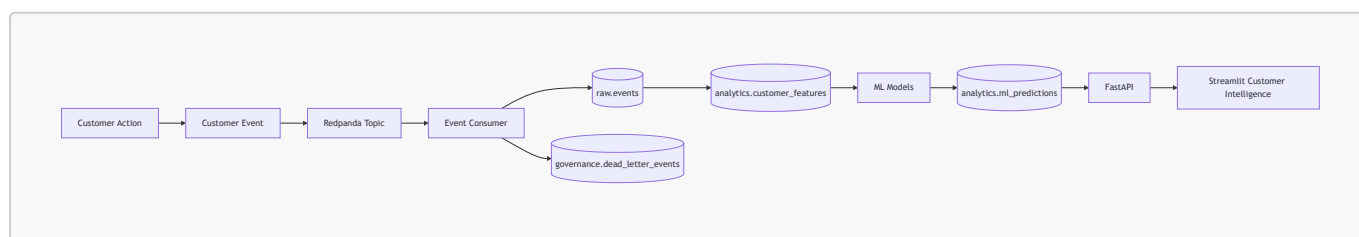
- PostgreSQL schemas;
- table names;
- column structures;
- model report files;
- Airflow DAGs;
- API endpoints;
- dashboard pages.

21.2 Business metadata

Business metadata exists through:

- glossary definitions;
- role ownership;
- retention policies;
- quality rule names;
- segment labels;
- model labels;
- dashboard descriptions.

21.3 Lineage overview



21.4 Future lineage improvements

Future improvements include:

- formal data catalog;
- automated lineage extraction;
- data asset owner registry;
- metadata tables;
- dbt documentation;
- OpenLineage integration.

22. Auditability and Evidence

Auditability is a core part of the RetailFlow governance framework.

Governance decisions and technical controls must produce evidence.

22.1 Audit evidence sources

Evidence source	Purpose
<code>governance.retention_actions_log</code>	Proves retention and anonymization actions.
<code>governance.dead_letter_events</code>	Proves invalid event isolation.
<code>governance.data_quality_logs</code>	Proves quality rule execution and failures.
Airflow DAG logs	Prove scheduled workflow execution.
ML reports	Prove model validation and monitoring.
Prometheus metrics	Prove platform monitoring.
Grafana dashboards	Visualize operational health.
Streamlit governance page	Provides governance visibility.
Streamlit Customer Intelligence page	Proves that customer-level AI predictions are blocked when analytics consent is absent.
Streamlit AI Monitoring page	Proves that AI-authorized prediction visibility is aligned with <code>analytics_consent_count</code> .
Streamlit Project Evidence page	Provides a final evidence matrix and skills evidence matrix for traceability to evaluation criteria.
Prometheus alert rules	Prove operational monitoring controls for service availability, latency, error rate and PostgreSQL status.
GitHub Actions reports	Prove CI/CD validation, tests and automated security checks.

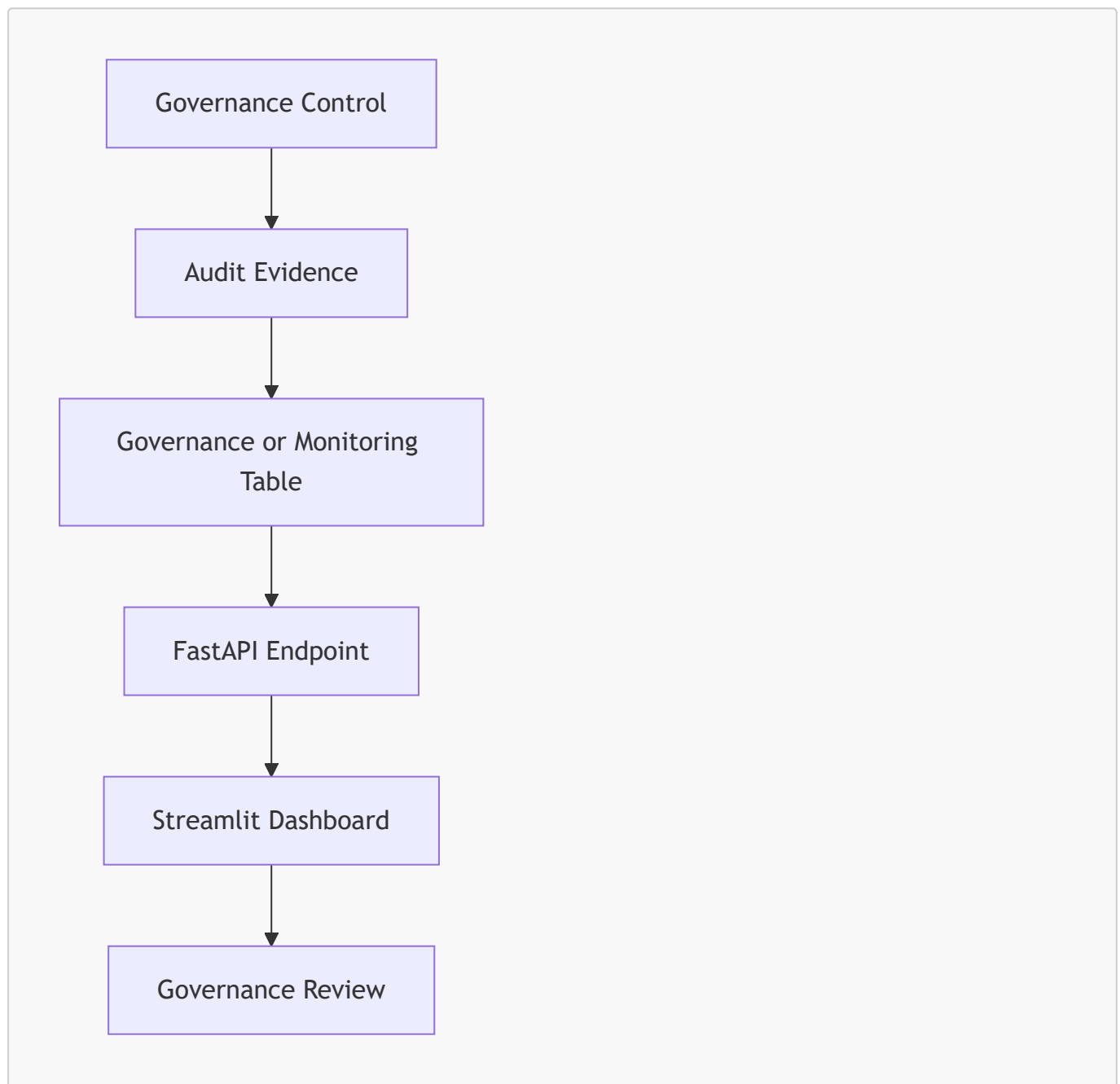
22.2 Audit questions RetailFlow can answer

RetailFlow can answer governance questions such as:

- Which customers have analytics consent?
- Which retention policies exist?
- Which customer records were anonymized?

- When was an anonymization action executed?
- Which component executed the action?
- Which events were rejected?
- Why were events rejected?
- Which quality rule failed?
- Are AI models monitored?
- Has drift been detected?
- Are platform services healthy?
- Are AI predictions hidden for customers without analytics consent?
- How many customers are authorized for AI visibility according to `analytics_consent_count`?
- Which implemented page or tool proves each academic skill or block requirement?

22.3 Audit flow



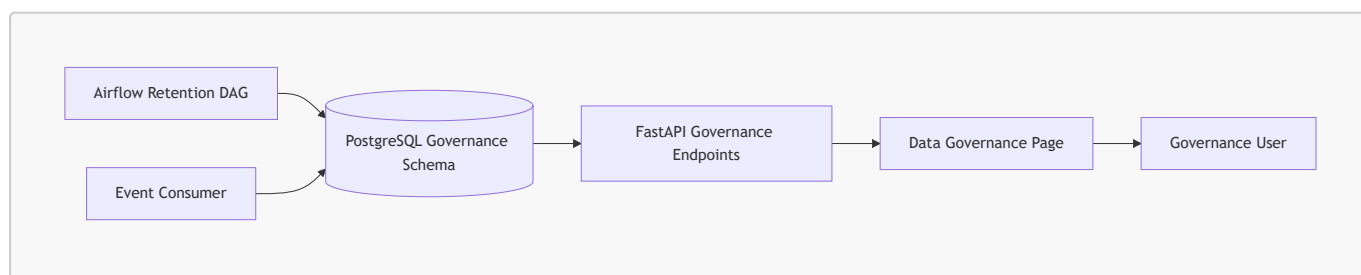
23. Technology and Tooling

RetailFlow uses technical tools to make governance operational.

23.1 Governance technology map

Capability	Tool / component
Consent storage	PostgreSQL customer and governance tables
Retention policy storage	<code>governance.data_retention_policies</code>
Retention automation	Airflow <code>retention_cleanup</code> DAG
Anonymization audit	<code>governance.retention_actions_log</code>
Event validation	Python event consumer validators
Dead-letter handling	<code>governance.dead_letter_events</code>
Quality logs	<code>governance.data_quality_logs</code>
Governance API	FastAPI <code>/governance/*</code> endpoints
Governance dashboard	Streamlit Data Governance page
Quality dashboard	Streamlit Data Quality page
AI monitoring	Streamlit AI Monitoring page
Project evidence traceability	Streamlit Project Evidence page with final evidence matrix and skills evidence matrix
Operational monitoring	Prometheus alert rules and Grafana dashboards
CI/CD governance evidence	GitHub Actions tests and security reports
Database access control evidence	PostgreSQL read-only role
Backup and restore evidence	PostgreSQL backup and restore scripts
Workflow orchestration	Airflow

23.2 Governance dashboard architecture



24. Governance KPIs

I defined governance KPIs to make governance measurable.

24.1 KPI table

KPI	Definition	Target	Owner
Analytics consent coverage	Share of customers with analytics consent enabled.	$\geq 75\%$	DPO / Compliance Lead
AI-authorized customer count	Number of customers whose AI outputs may be displayed according to <code>analytics_consent_count</code> .	Monitored and explained	DPO / Compliance Lead and ML Owner
Marketing consent coverage	Share of customers with marketing consent enabled.	$\geq 50\%$	Business Owner
Personalization consent coverage	Share of customers with personalization consent enabled.	$\geq 50\%$	Business Owner
Retention policy coverage	Share of critical governed domains covered by a retention policy.	100% for critical domains	DPO / Compliance Lead
Retention action traceability	Share of executed retention actions recorded in the audit log.	100%	Data Custodian
Data quality execution rate	Share of scheduled quality checks executed successfully.	$\geq 95\%$	Data Steward
Dead-letter rate	Share of rejected events among ingested events.	$< 2\%$	Data Steward
High severity issue resolution time	Average time to resolve high severity quality issues.	< 5 business days	Data Steward
ML monitoring coverage	Share of production ML models with metrics and drift monitoring.	100%	ML Owner
Evidence coverage	Share of major academic criteria mapped to an implemented proof.	100% for defended scope	Governance Council
Governance review cadence	Share of planned governance reviews completed.	$\geq 90\%$	Governance Council

24.2 KPI dashboard logic

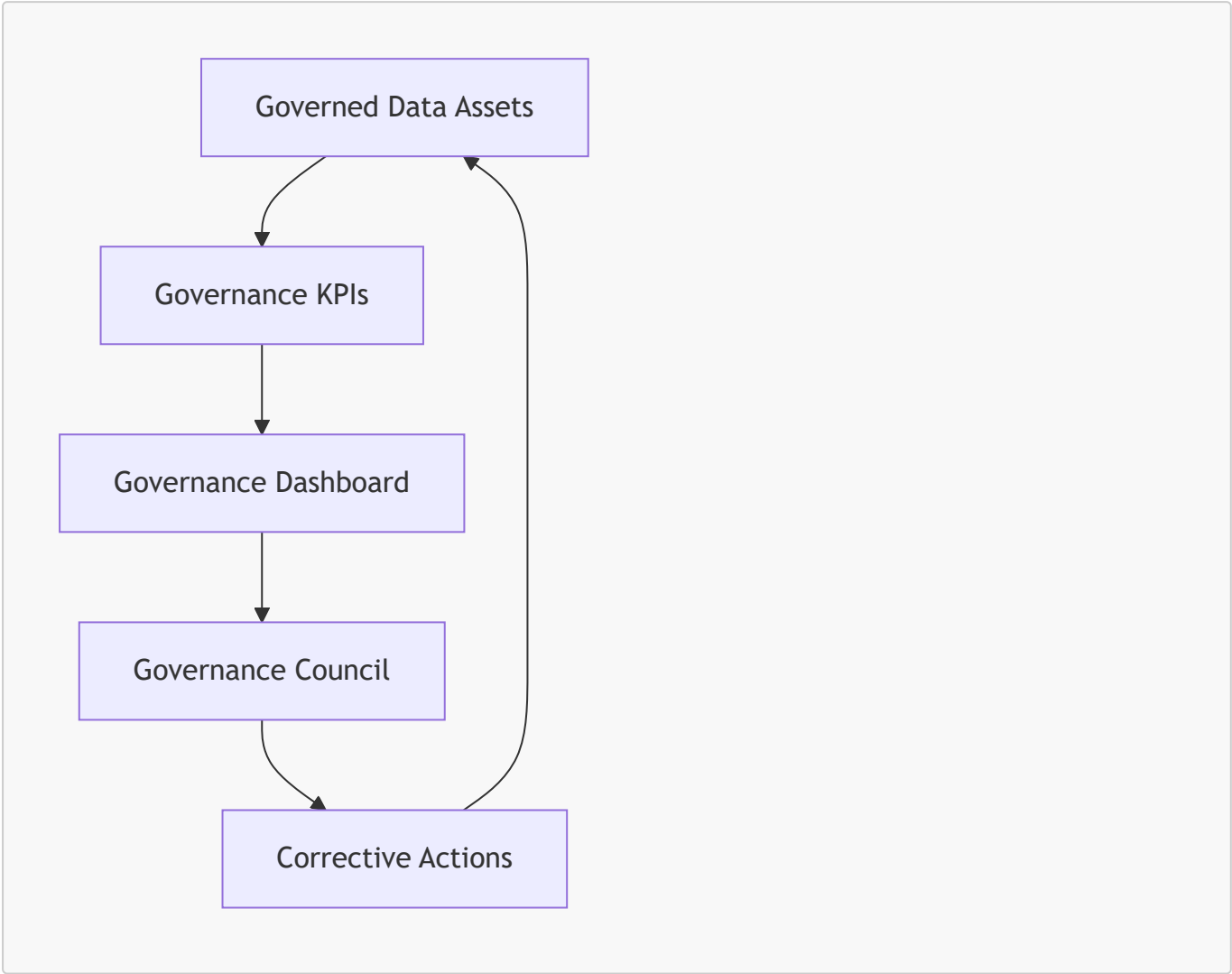
Governance KPIs should be reviewed monthly.

They should be presented to the Governance Council with:

- current value;
- target;
- trend;
- owner;

- issue explanation;
- corrective actions.

24.3 KPI diagram



25. Risk Management

I defined a governance risk register to identify and mitigate data risks.

25.1 Risk register

Risk	Description	Impact	Mitigation	Owner
Personal data exposure	Customer data may be accessed or used beyond intended purpose.	High	Consent management, classification, access principles, anonymization	DPO / Compliance Lead
Consent misuse	Customer data may be used for analytics or marketing without appropriate consent.	High	Consent-aware dashboards, AI display blocking when analytics consent is absent and policy review	DPO / Compliance Lead

Risk	Description	Impact	Mitigation	Owner
Retention failure	Data may be kept longer than needed.	High	Retention policies, Airflow cleanup, audit logs	Data Custodian
Poor data quality	Invalid events may affect analytics and ML models.	High	Validation rules, quality logs, dead-letter handling	Data Steward
ML drift	Customer behavior may change and reduce model reliability.	Medium	Drift monitoring and retraining DAG	ML Owner
Unclear ownership	Issues may remain unresolved if accountability is unclear.	Medium	Operating model and role mapping	Governance Council
Dashboard misuse	Users may overinterpret AI outputs.	Medium	Training, metric guides, human oversight and business-readable governance messages	Business Owner
Metadata gaps	Users may misunderstand data meaning or lineage.	Medium	Glossary and future catalog roadmap	Data Steward
Accessibility gap	Some users may not be able to follow standard training formats.	Medium	Accessible materials and multi-format training	Executive Sponsor

25.2 Risk review process

Risks should be reviewed quarterly by the Governance Council.

For each risk, the council should review:

- current likelihood;
- current impact;
- controls in place;
- control effectiveness;
- open actions;
- owner;
- deadline.

26. Inclusion and Accessibility

I included inclusion and accessibility in the governance framework because governance only works if users understand and can apply it.

Training and adoption cannot assume that every employee has the same language, learning style, availability or accessibility needs.

26.1 Inclusion principles

I defined the following principles:

1. Governance training should be understandable for both technical and non-technical users.
2. Training should be available in multiple languages when teams are international.
3. Documentation should avoid unnecessary jargon.
4. Important policies should be available in accessible formats.
5. Alternative training formats should be provided when needed.
6. Reasonable accommodations should be planned for people with disabilities.
7. Governance adoption should be measured without excluding users who need adapted support.

26.2 Multi-language training

RetailFlow should provide governance awareness material in multiple languages when required by the organization.

Priority languages should be based on workforce composition.

Training should cover:

- data ownership;
- consent usage;
- data quality responsibilities;
- retention and anonymization;
- AI output interpretation;
- incident reporting.

26.3 Accessibility accommodations

Training and documentation should support:

- screen-reader compatible documents;
- captions for recorded training;
- clear visual contrast;
- readable font sizes;
- non-visual alternatives for diagrams;
- extra time or assisted sessions when needed;
- simplified summaries for non-specialist audiences.

26.4 Inclusion in change management

Inclusion must not be treated as a separate afterthought.

It should be integrated into the governance change management plan.

This ensures that governance adoption is fair, accessible and realistic.

27. Change Management

Governance succeeds only if people adopt it.

I therefore included a change management plan.

27.1 Change management objectives

The change management plan aims to:

- explain why governance matters;
- clarify responsibilities;
- train data users;
- reduce resistance;
- make governance part of daily work;
- support inclusion and accessibility;
- create feedback loops.

27.2 Stakeholder communication

Audience	Message
Executive Sponsor	Governance protects data value, trust and platform scalability.
Data Owners	Governance clarifies accountability and improves decision quality.
Data Stewards	Governance gives structure to quality and issue resolution.
Data Custodians	Governance requirements are translated into technical controls.
Business Users	Governance makes dashboards more trustworthy and usable.
ML Users	Governance improves responsible use of predictions.

27.3 Training plan

Training module	Audience	Format
Governance fundamentals	All data users	Short online module
Consent and privacy	Marketing, analytics, business users	Workshop and quick reference sheet
Data quality issue reporting	Data users and stewards	Practical session
AI output interpretation	Business and ML users	Dashboard walkthrough
Retention and anonymization	DPO, data custodians, stewards	Process training
Accessibility and inclusion	Managers and trainers	Awareness session

27.4 Adoption strategy

I defined the following adoption strategy:

1. Start with customer and event domains.
2. Train the most active data users first.

3. Use dashboards to make governance visible.
4. Review KPIs monthly.
5. Collect feedback from users.
6. Improve policies based on recurring issues.
7. Extend governance to additional domains.

28. Implementation Roadmap

RetailFlow already includes several governance components.

The roadmap therefore includes both completed work and future improvements.

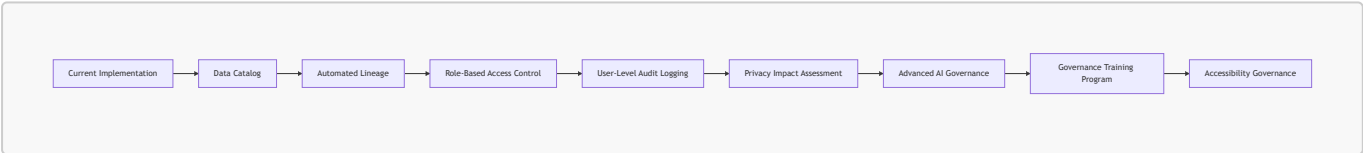
28.1 Completed governance implementation

Area	Completed implementation
Consent management	Customer consent fields and consent dashboard
Analytics consent	Consent-aware customer explorer and AI display blocking when analytics consent is absent
Retention policies	Retention policy table
Retention workflow	Airflow <code>retention_cleanup</code> DAG
Anonymization	Customer anonymization logic
Audit trail	Retention action log
Data quality	Validation rules and quality logs
Dead-letter handling	Invalid events stored in governance dead-letter table
Governance dashboard	Streamlit Data Governance page
Data Quality dashboard	Streamlit Data Quality page
AI governance	AI monitoring, drift, model registry, retraining evidence and consent-aware AI visibility
Project evidence	Final evidence matrix and skills evidence matrix mapped to blocks and skills
Observability evidence	Prometheus alert rules, Grafana dashboards and Streamlit Observability page
CI/CD governance evidence	GitHub Actions tests, Docker validation and security reports
Operational resilience evidence	Healthchecks, backup/restore scripts and PostgreSQL read-only role
Orchestration	Airflow governance and ML workflows

28.2 Future roadmap

Phase	Improvement	Purpose
Phase 1	Formal data catalog	Improve discoverability and ownership visibility.
Phase 2	Automated lineage	Trace data from source events to dashboards and ML outputs.
Phase 3	Role-based access control	Strengthen access governance by persona.
Phase 4	User-level audit logging	Track who accessed sensitive datasets or dashboards.
Phase 5	Advanced privacy impact assessment	Formalize privacy review for new use cases.
Phase 6	Extended AI governance	Add fairness checks, model registry and approval workflow.
Phase 7	Governance training program	Institutionalize awareness and adoption.
Phase 8	Accessibility governance	Ensure training and documentation remain inclusive.

28.3 Roadmap diagram



29. Future Improvements

The governance framework is already strong for the current stage of RetailFlow, but I identified several future improvements.

29.1 Data catalog

A data catalog would make assets easier to find and understand.

It should include:

- table descriptions;
- column descriptions;
- owner;
- steward;
- classification;
- retention policy;
- glossary links;
- quality rules;

- lineage.

29.2 Metadata automation

Metadata is currently documented through schemas, code and dashboards.

A future improvement would automate metadata collection.

29.3 Advanced lineage

Lineage should be extended to show:

```
source event
→ raw table
→ feature table
→ ML prediction
→ API endpoint
→ Streamlit dashboard
```

29.4 Role-based access control

Future platform versions should enforce access based on roles such as:

- Data Steward;
- Business User;
- ML Engineer;
- Compliance Lead;
- Platform Engineer.

29.5 Model registry and AI approval

The AI governance framework can be strengthened with:

- model registry;
- approval workflow;
- deployment stages;
- rollback strategy;
- production validation checklist.

29.6 Governance issue tracker

A dedicated governance issue tracker would help manage:

- quality issues;
- glossary updates;
- policy questions;
- access requests;
- privacy reviews.

29.7 Accessibility review process

Accessibility should be reviewed periodically for:

- training content;
 - dashboards;
 - Project Evidence and Skills Evidence matrices;
 - AI Monitoring metrics aligned with `analytics_consent_count`;
 - Prometheus alert rules, Grafana dashboards and CI/CD evidence;
 - documentation;
 - diagrams;
 - onboarding material.
-

30. Conclusion

I designed the RetailFlow data governance framework as an operational component of the platform, not as a separate theoretical layer.

The framework covers ownership, policies, consent, retention, anonymization, quality, AI monitoring, auditability, inclusion and continuous improvement.

The main strength of the approach is that governance is embedded directly into the platform design and visible in the final application experience.

This is visible through:

- consent-aware customer intelligence with AI prediction blocking when analytics consent is absent;
- retention policies and anonymization workflow;
- dead-letter event handling;
- quality logs;
- governance APIs;
- dashboards;
- Project Evidence and Skills Evidence matrices;
- AI Monitoring metrics aligned with `analytics_consent_count`;
- Prometheus alert rules, Grafana dashboards and CI/CD evidence;
- Airflow workflows;
- AI monitoring;
- risk and KPI management.

RetailFlow therefore demonstrates a mature approach to data governance for a modern Retail Intelligence platform.

The governance framework makes customer data more trustworthy, business intelligence more reliable, AI outputs more responsible and platform operations more auditable.